AI-Based Smart Agricultural Decision Support System using Deep Learning, Machine Learning and Agentic AI

A Mini Project Report
submitted in partial fulfillment of the requirements for
the award of the degree of

Bachelor of Engineering

in

Artificial Intelligence and Data Science

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HYDERABAD – 500075
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2. Develop professional skills in the thrust areas like ANN and Deep learning, Robotics, Internet of Things and Big Data Analytics.
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1. **Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
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3. **Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
4. **Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
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6. **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
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9. **Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
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11. **Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)



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MINI PROJECT-II

COURSE OBJECTIVES

1. To enable students learning by doing.
2. To develop capability to analyse and solve real world problems.
3. To inculcate innovative ideas of the students.
4. To impart team building and management skills among students.
5. To instill writing and presentation skills for completing the project.

COURSE OUTCOMES

Upon successful completion of this course, students will be able to:

1. Interpret Literature with the purpose of formulating a project proposal.
2. Plan, Analyse, Design and Implement a project.
3. Find the solution of identified problem with the help of modern Technology and give priority to real time scenarios.
4. Plan to work as a team and to focus on getting a working project done and submit a report within a stipulated period of time.
5. Prepare and submit the Report and deliver presentation before the Departmental Committee.

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CO-PO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	3	3	3	3	2	1	2	3
CO2	3	3	3	3	3	3	3	2	1	2	3
CO3	3	3	3	3	3	3	3	2	-	2	3
CO4	2	2	2	3	3	3	3	2	3	3	2
CO5	1	2	1	2	3	3	-	-	2	3	-

Mapping of Course Outcomes with Program Outcomes

CO-PSO MAPPING

	PSO1	PSO2	PSO3
CO1	2	3	2
CO2	1	3	2
CO3	3	3	2
CO4	2	3	2
CO5	2	3	-

Mapping of Course Outcomes with Program Specific Outcomes

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DECLARATION CERTIFICATE

We hereby declare that the mini project titled **AI-Based Smart Agricultural Decision Support System using Deep Learning, Machine Learning and Agentic AI** submitted by us to the **Artificial Intelligence and Data Science CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD** in partial fulfillment of the requirements for the award of **Bachelor of Engineering** is a bona-fide record of the work carried out by us under the supervision of **Dr. Satya Kiranmai Tadepalli**. We further declare that the work reported in this project, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or University.

Project Associates

Sunkara Madhuchandana (1601-24-771-016)

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BONAFIDE CERTIFICATE

This is to certify that the project titled **AI-Based Smart Agricultural Decision Support System using Deep Learning, Machine Learning and Agentic AI** is a bonafide record of the work done by

Sunkara Madhuchandana (1601-24-771-016)

Sirigiri Lavanya (1601-24-771-022)

Thumu Madhuri (1601-24-771-025)

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Artificial Intelligence and Data Science** to the **CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD** carried out under my guidance and supervision during the year 2026-27. The results presented in this project report have not been submitted to any other university or Institute for the award of any degree.

Dr.Satya Kiranmai Tadepalli
Guide

Prof.K.Radhika
Head of the Department

Submitted for Semester Mini-Project viva-voce examination held on

_____

Examiner-1

Examiner-2

ABSTRACT

Agriculture is central to food security and sustainable development, yet farmers frequently struggle to choose suitable crops, estimate expected yield and plan irrigation under changing land and environmental conditions. This project presents an AI-Based Smart Agricultural Decision Support System that integrates Deep Learning, Machine Learning and Agentic AI to give farmers intelligent, data-driven recommendations through a single interface.

The system accepts a land image, location, land area, soil type and water availability as inputs. A MobileNetV3 model, fine-tuned on the EuroSAT RGB land-cover dataset, analyses the land image to decide whether the land is suitable for agriculture and to identify its condition as dry, wet, moist or vegetated. Temperature, humidity and rainfall are retrieved automatically for the selected location. Three Random Forest models then perform crop recommendation, crop yield prediction and irrigation recommendation. A single AI Agent acts as the coordinator: it invokes the required models in the correct order, combines their outputs and produces a final report containing land suitability analysis, the recommended crop, estimated yield, irrigation requirement and a plain-language explanation. The work aims to improve farming decisions, optimise the use of land and water, and support sustainable agriculture. Future enhancements include IoT sensor integration, satellite imagery, plant disease detection and market price prediction.

Keywords : Smart Agriculture, Decision Support System, MobileNetV3, EuroSAT, Random Forest, Agentic AI, Crop Recommendation, Yield Prediction, Irrigation Recommendation

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SUSTAINABLE DEVELOPMENT GOALS (SDGs)

SDG1: **No Poverty:** End poverty in all its forms everywhere.

SDG2: **Zero Hunger:** End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.

SDG3: **Good Health and Well-being:** Ensure healthy lives and promote well-being for all at all ages.

SDG4: **Quality Education:** Ensure inclusive and equitable quality education and promote lifelong learning opportunities.

SDG5: **Gender Equality:** Achieve gender equality and empower all women and girls.

SDG6: **Clean Water and Sanitation:** Ensure availability and sustainable management of water and sanitation.

SDG7: **Affordable and Clean Energy:** Ensure access to affordable, reliable, sustainable, and modern energy for all.

SDG8: **Decent Work and Economic Growth:** Promote sustained, inclusive, and sustainable economic growth, full and productive employment.

SDG9: **Industry, Innovation and Infrastructure:** Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

SDG10: **Reduced Inequality:** Reduce inequality within and among countries.

SDG11: **Sustainable Cities and Communities:** Make cities and human settlements inclusive, safe, resilient, and sustainable.

SDG12: **Responsible Consumption and Production:** Ensure sustainable consumption and production patterns.

SDG13: **Climate Action:** Take urgent action to combat climate change and its impacts.

SDG14: **Life Below Water:** Conserve and sustainably use the oceans, seas, and marine resources.

SDG15: **Life on Land:** Protect, restore, and promote sustainable use of terrestrial ecosystems.

SDG16: **Peace, Justice and Strong Institutions:** Promote peaceful and inclusive societies, provide access to justice, and build effective institutions.

SDG17: **Partnerships for the Goals:** Strengthen the means of implementation and revitalize global partnerships for sustainable development.



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PROJECT – SDG MAPPING

SDG	Relevance of the project
SDG2	Supports food security and sustainable agriculture by giving farmers data-driven crop recommendations and yield estimates that help them make better use of the land they have.
SDG6	The irrigation model estimates the water requirement (None, Low, Medium or High) for the recommended crop, encouraging efficient and sustainable use of water.
SDG9	Brings AI-based innovation to agriculture through an integrated, modular decision-support system that farmers can use through a simple web interface.
SDG12	Checks land suitability before recommending a crop and tailors crop and irrigation advice to soil, weather and water availability, reducing wasteful use of land and water.
SDG13	Retrieves current temperature, humidity and rainfall for the farmer's location so that crop and irrigation decisions adapt to changing weather conditions.
SDG15	The land analyser marks forest cover and water bodies as not suitable for cultivation, discouraging the conversion of natural ecosystems into farmland.

Mapping of the Project with Sustainable Development Goals

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CHAPTER 1
INTRODUCTION

Agriculture remains one of the most important sectors for ensuring food security and sustainable development. Every season, farmers must take a series of interrelated decisions: whether a piece of land is fit for cultivation, which crop will perform best on it, how much harvest can reasonably be expected, and how much water the crop will need. These decisions are traditionally made from experience, local advice or trial and error, and they become harder as weather patterns, soil conditions and water availability change from year to year.

This project proposes an AI-Based Smart Agricultural Decision Support System that brings these decisions together in one place. The system combines three families of artificial intelligence: Deep Learning, to understand the land from an image; Machine Learning, to predict the crop, yield and irrigation need from tabular agronomic and weather data; and Agentic AI, to coordinate the individual models and explain the final recommendation in language a farmer can follow.

1.1 Background

Smart farming applies data, sensing and analytics to make agricultural production more precise and less wasteful. Machine learning has been used across the farming cycle, including crop management, water management and soil management, while deep learning has proved particularly effective for image-based tasks such as land-cover mapping and plant disease recognition. Remote-sensing benchmarks such as EuroSAT, which contains labelled Sentinel-2 satellite patches of agricultural, natural and built-up land, make it possible to train compact models that recognise what kind of land is shown in an image.

Despite this progress, most available tools solve only one part of the problem. A crop recommendation model does not say whether the land itself is usable; a yield model does not say which crop to grow; and an irrigation model is rarely linked to the crop that was actually recommended. A farmer who wants a complete answer must run several tools separately and interpret numeric outputs without help. Recent developments in tool-using language-model agents offer a way to close this gap,

because an agent can decide which models to call, in which order, and how to present their combined results.

1.1.1 Problem Statement

Farmers lack a single, easy-to-use system that can (i) judge whether a given piece of land is suitable for agriculture and in what condition it is, (ii) recommend a suitable crop for the prevailing weather, soil and water situation, (iii) estimate the likely yield of that crop for the available land area, and (iv) advise on the irrigation required, together with a clear explanation of why each recommendation was made.

1.2 Objectives

The objectives of this project are:

1. To develop a deep learning model based on MobileNetV3 that analyses an uploaded land image and determines whether the land is suitable for agriculture and whether its condition is dry, wet, moist or vegetated.
2. To retrieve temperature, humidity and rainfall automatically for the location selected by the farmer.
3. To build Random Forest models for crop recommendation, crop yield prediction and irrigation recommendation using land, soil, water and weather inputs.
4. To design a single AI Agent that invokes the required models, combines their predictions and generates a final recommendation with an explanation.
5. To present the results through a simple interface that supports better farming decisions, efficient use of resources and sustainable agriculture.

1.3 Scope of the Project

The system accepts five inputs from the farmer: a land image, the location, the land area, the soil type and the water availability. It returns a land suitability analysis, a recommended crop, an estimated crop yield, an irrigation requirement and an explanation.

Within scope: image-based land suitability and condition analysis; automatic weather retrieval; crop recommendation, yield prediction and irrigation recommendation using Random Forest models; coordination of these components by a single agent; and a user-facing interface to present the final report.

Outside the current scope: real-time field sensing (IoT), live satellite imagery feeds, plant disease detection, market price prediction and multi-agent collaboration. These are identified as future work in Chapter 6.

Limitations to note. The land analysis model is trained on EuroSAT, which consists of top-down satellite imagery at 10 m resolution. The system therefore expects a top-down satellite or aerial image of the land rather than a ground-level photograph. Recommendations are decision support for the farmer and do not replace advice from local agricultural experts or a laboratory soil test.

1.3.1 Organisation of the Report

Chapter 2 reviews related work and the theory behind the models used. Chapter 3 describes data collection, analysis techniques and the design of the system. Chapter 4 presents the design specifications, implementation steps, tools and results. Chapter 5 analyses the findings and compares them with existing work. Chapter 6 concludes the report and outlines future work.

CHAPTER 2
LITERATURE REVIEW

This chapter surveys prior research on the use of deep learning, machine learning and intelligent agents in agriculture, explains the theory behind the techniques selected for this project, summarises the main findings from the literature and describes the methodology adopted.

2.1 Previous Work

Crop recommendation research commonly applies supervised machine-learning methods to relate soil and weather conditions to suitable crops. Sani et al. [3] study a Random Forest-based recommendation system, while ReenaSri et al. [6] examine machine-learning algorithms for crop recommendation. Sagar et al. [4] present Crop Connect as a crop-recommendation approach, and Tomar et al. [15] and Palei and Mohapatra [16] investigate machine-learning and deep-learning approaches for crop selection. Das and Chatterjee [5] specifically emphasise explainable machine learning using agricultural sensor data. Kumar and Kumar [18] further propose an XAI-based crop recommendation framework using soil nutrient profiles and historical crop yields.

Crop-yield prediction is another important direction in data-driven agriculture. Sakib et al. [1] investigate predictive analytics to identify determinants of crop yield using machine learning. Nurcahyo et al. [8] and [17] address interpretable machine learning for multi-class crop-yield prediction, while Khaki et al. [19] describe a CNN–RNN framework for crop-yield prediction. Mohan et al. [9] and [20] discuss integrating artificial intelligence and explainable AI for precision crop-yield prediction. These studies motivate combining yield estimates with crop-selection decisions, while recognising that the exact performance depends on the datasets, input features and evaluation protocols used.

Explainability is increasingly relevant because agricultural recommendations should be understandable to farmers and other decision-makers. Nikam et al. [2] present a LIME-based explainable approach for species identification. Das and Chatterjee [5] focus on explainable crop recommendation, and Kumar and Kumar [18] propose an XAI-based crop recommendation framework. Kisten et al. [7] apply explainable AI to predictive maintenance in smart agricultural facilities, while Chen et al. [10] discuss

combining explainable AI and blockchain for smart agriculture. Diachenko et al. [11] describe a decision-support system for wheat powdery mildew risk prediction using weather monitoring, machine learning and explainable AI. Work on broader agricultural optimisation and planning includes Bonthu et al. [12] and Murugan et al. [14], while Khaki et al. [13] investigate land-suitability assessment through fusion of Sentinel-1 and Sentinel-2 imagery. Taken together, the cited work covers crop recommendation, yield prediction, land suitability, agricultural risk prediction and explainability; the present project draws on these themes to motivate an integrated decision-support workflow.

2.2 Theoretical Framework

2.2.1 Convolutional Neural Networks and MobileNetV3

A convolutional neural network (CNN) learns hierarchical visual features through stacked convolution, non-linearity and pooling layers. MobileNet-family networks are designed for efficient inference on resource-limited devices. They build on depthwise separable convolutions, which split a standard convolution into a per-channel spatial filter and a 1 × 1 pointwise convolution to reduce computation, and on inverted residual blocks with linear bottlenecks. MobileNetV3 adds squeeze-and-excitation attention modules, a network architecture found through hardware-aware neural architecture search, and the hard-swish activation, defined as

$$\text{h-swish}(x) = x \cdot \frac{\text{ReLU6}(x+3)}{6} \tag{2.1}$$

Hard-swish approximates the smooth swish function using only operations that are cheap on mobile hardware. MobileNetV3 is offered in Large and Small variants; the Large variant is used here because it offers higher accuracy at a cost still low enough for a web application.

2.2.2 Transfer Learning

Instead of training a network from random weights, transfer learning starts from a model pretrained on a large dataset (ImageNet) and adapts it to the new task. First the pretrained backbone is frozen and only a new classification head is trained; then the top layers of the backbone are unfrozen and fine-tuned with a small learning rate. This reduces training time and the amount of data needed, and generally improves accuracy.

2.2.3 Random Forest

A Random Forest is an ensemble of decision trees, each grown on a bootstrap sample of the training data with a random subset of features considered at every split. For classification the forest predicts the class chosen by the majority of its T trees, and for regression it predicts the mean of the tree outputs:

$$\hat{y}_{\text{class}} = \text{mode}\{h_t(x)\}_{t=1}^{T}, \quad \hat{y}_{\text{reg}} = \frac{1}{T}\sum_{t=1}^{T} h_t(x) \tag{2.2}$$

Random Forests handle mixed numerical and categorical features, are resistant to over-fitting, need little tuning and provide feature-importance scores. The last property is valuable here because it supplies material for the explanations the agent gives the farmer.

2.2.4 Agentic AI

An AI agent is a system in which a language model is given a goal, a set of tools and the ability to decide which tool to call next based on what it has observed so far. The agent loop alternates between reasoning about the current state, acting by calling a tool, and observing the result. In this project the tools are the MobileNetV3 land analyser, the weather service and the three Random Forest models. A single agent is sufficient because the task is a well-defined sequence in which the output of one tool becomes the input of the next.

2.3 Key Findings

The review leads to the following observations that shape the present work:

1. Compact CNNs reach very high accuracy on EuroSAT, so a lightweight MobileNetV3 is a sound choice for land analysis.
2. Random Forests are strong, interpretable baselines for crop recommendation and yield prediction on tabular data.
3. Temperature, humidity, rainfall and soil type are the most informative inputs across crop and yield studies.
4. Existing systems usually address a single task (land classification, crop recommendation, yield prediction or irrigation) and leave the farmer to combine the results.

5. Few farmer-facing systems verify whether the land is suitable before recommending a crop, and few explain their recommendations.
6. Tool-using agents can coordinate several specialised models and produce a consolidated explanation.

The identified gap is an integrated, explainable system that begins with land suitability analysis and ends with crop, yield and irrigation advice, coordinated by an agent. This is the gap the present project addresses.

Work	Focus	Technique	Gap
Sakib et al. [1]	Crop-yield determinants	Predictive analytics and machine learning	Does not by itself integrate land suitability and irrigation advice
Sani et al. [3]	Crop recommendation	Random Forest	Focuses on crop selection
Das and Chatterjee [5]	Explainable crop recommendation	Explainable ML with agricultural sensor data	Scope centres on recommendation and sensor data
Nurcahyo et al. [8, 17]	Multi-class crop-yield prediction	Interpretable machine learning	Yield prediction focus
Khaki et al. [13]	Land suitability assessment	Sentinel-1 and Sentinel-2 feature-level fusion	Case study focuses on suitability mapping
Kumar and Kumar [18]	XAI-based crop recommendation	Soil nutrient profiles and historical yields	Recommendation framework focus
This project	Land suitability, crop, yield and irrigation advice	Deep learning, machine learning and agent orchestration	Integrated farmer-facing decision support

Table 2.1: Summary of related work and the gap addressed

2.4 Methodology

The project follows a staged, data-science methodology in the spirit of CRISP-DM, adapted to a system with several models and an orchestrating agent:

1. **Problem understanding:** define the farmer's decisions and the inputs and outputs of the system.
2. **Data collection and preparation:** gather EuroSAT RGB images, the tabular agronomic datasets and a weather source; clean, encode and split the data.
3. **Model development:** fine-tune MobileNetV3 for land analysis, and train and tune three Random Forest models.

4. **Agent design:** define the tools, their input and output contracts, the order of calls and the explanation format.
5. **Integration and interface:** connect the agent, models and weather service to a web interface.
6. **Evaluation:** measure each model with suitable metrics and test the full system on representative scenarios.

CHAPTER 3
RESEARCH DESIGN

This chapter explains how data were collected, how they were analysed and how the overall system was designed.

3.1 Data Collection Methods

The system draws on four sources of data, summarised in Table 3.1.

Data	Content	Used by	Source
EuroSAT RGB	27,000 labelled satellite image patches, 10 classes	MobileNetV3 land analyser	Public benchmark
Crop recommendation data	Soil, weather and water attributes mapped to suitable crops	Crop recommendation model	Public agronomic datasets
Yield and irrigation data	Crop, soil, area, weather, water availability, yield and irrigation level	Yield and irrigation models	Public agricultural records, FAO guidelines
Weather	Temperature, humidity and rainfall for a location	Agent and all Random Forest models	Weather API

Table 3.1: Data sources used in the project

3.1.1 EuroSAT RGB Dataset

EuroSAT is built from Sentinel-2 satellite images covering cities in 34 European countries. The RGB version used here contains 27,000 labelled and geo-referenced patches of 64 × 64 pixels in the visible red, green and blue bands, at a ground sampling distance of 10 m, so that each patch covers roughly 640 m × 640 m of land. The ten classes and their sizes are listed in Table 3.2.

No.	Class	Images	No.	Class	Images
1	Annual Crop	3,000	6	Pasture	2,000
2	Forest	3,000	7	Permanent Crop	2,500
3	Herbaceous Vegetation	3,000	8	Residential	3,000
4	Highway	2,500	9	River	2,500
5	Industrial	2,500	10	Sea & Lake	3,000
Total: 27,000 images					

Table 3.2: EuroSAT RGB class distribution

3.1.2 From Land-Cover Classes to Suitability and Condition

EuroSAT labels describe land cover, whereas the system must report suitability for agriculture and land condition. The CNN therefore predicts one of the ten EuroSAT classes, and a mapping layer converts this class into a suitability decision and a condition label, as shown in Table 3.3. Classes that are cultivated or naturally vegetated and usable for farming are marked suitable. Built-up land, water bodies and forest are marked not suitable because they cannot be put under crops directly. Water classes give the condition wet and vegetation classes give vegetated. For annual cropland, whose surface can be bare or growing, the condition is resolved as dry or moist using a rule on the humidity and recent rainfall obtained from the weather service (the thresholds are configurable parameters of the system).

EuroSAT Class	Suitable?	Land Condition
Annual Crop	Yes	Dry or Moist (weather rule)
Permanent Crop	Yes	Vegetated
Pasture	Yes	Vegetated
Herbaceous Vegetation	Yes	Vegetated
Forest	No (forest cover)	Vegetated
River	No (water body)	Wet
Sea & Lake	No (water body)	Wet
Highway	No (built-up)	Dry (non-arable)
Industrial	No (built-up)	Dry (non-arable)
Residential	No (built-up)	Dry (non-arable)

Table 3.3: Mapping of EuroSAT classes to land suitability and condition

3.1.3 Tabular Agronomic Data

Three tabular datasets support the Random Forest models. Their input and output variables are summarised in Table 3.4. Soil type and water availability are categorical inputs provided by the farmer, while temperature, humidity and rainfall are numeric and come from the weather service.

Model	Input features	Output
Crop recommendation (classifier)	Soil type, temperature, humidity, rainfall, water availability, land condition	Recommended crop
Yield prediction (regressor)	Crop, soil type, land area, temperature, humidity, rainfall, water availability	Yield per hectare, scaled to total yield for the land area
Irrigation recommendation (classifier)	Crop, soil type, temperature, humidity, rainfall, water availability	Irrigation requirement: None, Low, Medium or High

Table 3.4: Input and output variables of the Random Forest models

3.1.4 Weather Data

When the farmer selects a location, the system converts it to geographic coordinates and calls a public weather API to obtain the current temperature (°C), relative humidity (%) and rainfall (mm). No manual weather entry is needed, which removes a common source of input error.

3.1.5 Farmer Inputs

At run time the farmer supplies the land image, location, land area, soil type and water availability. The image is validated (format and size), the area is checked for a positive value, and soil type and water availability are chosen from fixed lists to keep them consistent with the training data.

3.2 Data Analysis Techniques

3.2.1 Image Preprocessing and Augmentation

EuroSAT images are resized to 224 × 224 pixels to match the input size of the pretrained MobileNetV3 backbone. The dataset is divided into 70% training, 15% validation and 15% test images using a fixed random seed. Augmentation is applied only to the training set and consists of random horizontal and vertical flips and small random rotations, which suit top-down imagery whose orientation has no fixed meaning.

3.2.2 Tabular Data Preparation

The tabular datasets are examined for missing values, duplicates and outliers. Categorical features (soil type, water availability, land condition, crop) are one-hot encoded; numeric features are used as they are because tree-based models do not need scaling. Each dataset is split into training and test sets, and hyper-parameters are chosen by cross-validated grid search on the training portion only.

3.2.3 Model Training

MobileNetV3. Training is done in two phases. In the first phase the pretrained backbone is frozen and only the new classification head is trained. In the second phase the upper blocks of the backbone are unfrozen and fine-tuned with a much lower learning rate. Early stopping on validation loss prevents over-fitting.

Random Forests. Each of the three models is a Random Forest trained with scikit-learn[24]. The number of trees, maximum depth and minimum samples per split are tuned by grid search with cross-validation. Feature-importance scores are stored with each model for use in explanations.

3.2.4 Evaluation Metrics

The classifiers (MobileNetV3, crop recommendation and irrigation) are evaluated with accuracy, precision, recall and F1-score:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{3.1}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.2}$$

For multi-class problems, macro-averaged values are reported, and a confusion matrix shows per-class behaviour. The yield regressor is evaluated with the root mean squared error, mean absolute error and coefficient of determination:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

3.3 Project Design and Development

3.3.1 System Architecture

The system is organised around a single AI Agent, as shown in Figure 3.1. The farmer's inputs go to the agent, which calls five tools: the MobileNetV3 land analyser, the weather service, and the three Random Forest models. The agent gathers the tool outputs and returns one consolidated recommendation with an explanation.

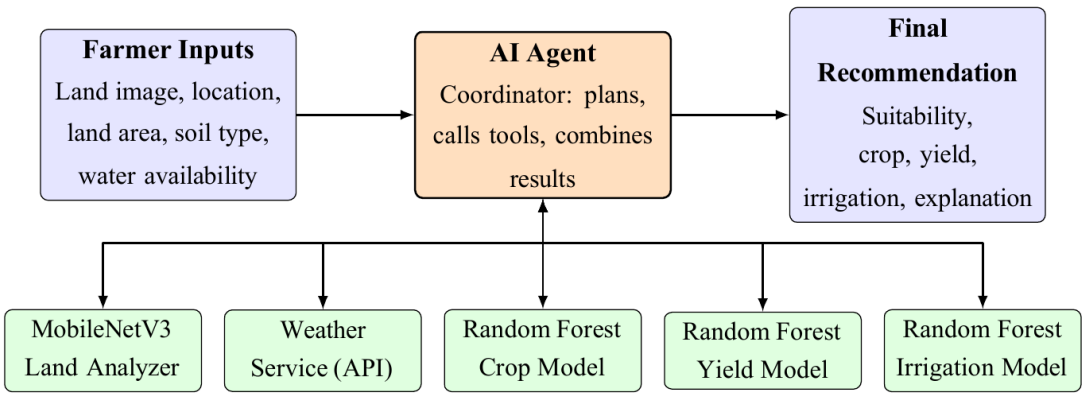


Figure 3.1: Architecture of the AI-Based Smart Agricultural Decision Support System

3.3.2 Module Description

Module / Tool	Input	Output
Land Analyzer (MobileNetV3)	Land image	Land-cover class, suitability (yes/no), land condition
Weather Service	Location	Temperature, humidity, rainfall
Crop Recommender (RF)	Soil type, weather, water availability, land condition	Recommended crop
Yield Predictor (RF)	Recommended crop, soil type, land area, weather, water availability	Yield per hectare and total estimated yield
Irrigation Model (RF)	Recommended crop, soil type, weather, water availability	Irrigation requirement level
AI Agent	Farmer inputs and all tool outputs	Final recommendation report with explanation

Table 3.5: Modules of the system and their interfaces

3.3.3 AI Agent Design

The agent is the only component that talks to the other modules, which keeps the design simple and each model independent and replaceable. Its responsibilities are:

1. **Planning:** decide which tools are needed and the order of calls, so that each tool receives the outputs it depends on. For example, the yield and irrigation models need the crop chosen by the crop recommender.
2. **Tool invocation:** call each tool with correctly formatted inputs and read its output.
3. **Conditional control:** if the land analyser reports that the land is not suitable for agriculture, stop the pipeline and explain why, rather than recommending a crop for unusable land.

4. **Combination and explanation:** merge the outputs into a single report and explain each recommendation using the input values, the weather retrieved and the most influential features reported by the Random Forest models.
5. **Grounding:** all numbers in the report (yield, irrigation level, weather) are taken from tool outputs and never generated freely by the language model, which prevents invented values.

3.3.4 Workflow

The end-to-end workflow is shown in Figure 3.2.

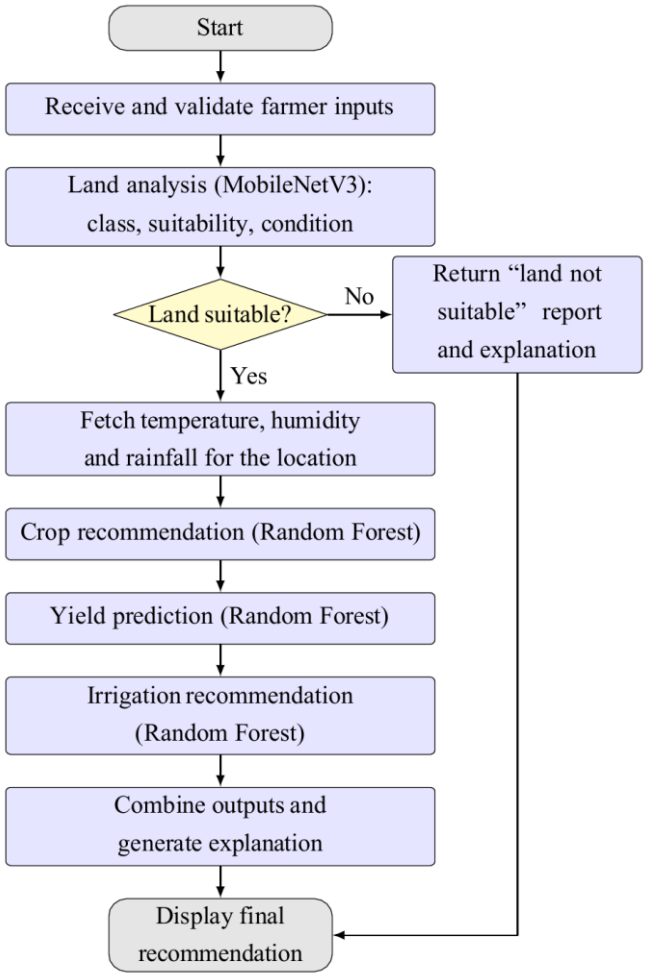


Figure 3.2: Workflow of the agent-coordinated recommendation process

3.3.5 Output Report

The final report shown to the farmer contains five parts: (i) the land suitability analysis with the land condition, (ii) the recommended crop, (iii) the estimated yield for the given land area, (iv) the irrigation requirement, and (v) an explanation that relates

each recommendation to the soil type, water availability, weather and land condition supplied or retrieved.

CHAPTER 4
DESIGN SPECIFICATIONS

This chapter describes how the system was implemented, the tools and technologies used, and the results obtained from testing.

4.1 Implementation Steps

4.1.1 Step 1: Land Analysis Model

The EuroSAT RGB images are loaded from class-labelled folders, resized to 224 × 224 and divided into training, validation and test sets. A MobileNetV3-Large backbone pretrained on ImageNet is used without its original classification layer. A global average pooling layer, a fully connected layer with dropout and a ten-way softmax layer are added on top (Figure 4.1). After prediction, the class-to-suitability mapping of Table 3.3 yields the suitability and condition of the land.

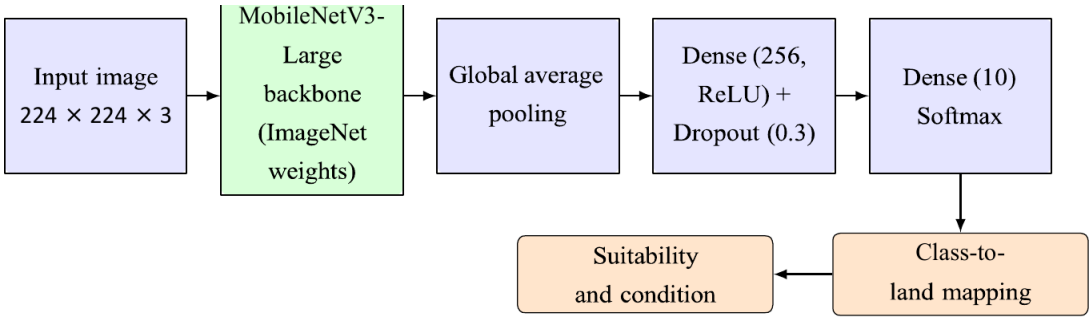


Figure 4.1: MobileNetV3 transfer-learning pipeline for land analysis

4.1.2 Step 2: Weather Retrieval

A function takes the location name, finds its coordinates through a geocoding request and then requests the current temperature, relative humidity and rainfall from the weather API. The result is returned as a small structured record. Network errors and unknown locations are caught and reported to the agent, which asks the farmer to correct the location.

4.1.3 Step 3: Random Forest Models

The crop recommendation, yield prediction and irrigation datasets are cleaned and encoded as described in Section 3.2. Each Random Forest is built as a scikit-learn pipeline that combines the one-hot encoder and the forest, so that the same preprocessing is applied at training and prediction time. The trained pipelines are saved

to disk with joblib and loaded once when the application starts. The total expected yield is obtained by multiplying the predicted yield per hectare by the land area expressed in hectares.

4.1.4 Step 4: Agent Tools and Orchestration

Each model is wrapped as a tool with a name, a short description and a typed input and output schema: `analyze_land`, `get_weather`, `recommend_crop`, `predict_yield` and `recommend_irrigation`. The agent receives the farmer's inputs and a system instruction describing the order of calls, the stopping rule for unsuitable land and the requirement that all numbers come from tool outputs. After the tools have run, the agent writes the explanation using the tool outputs and the stored feature-importance information.

4.1.5 Step 5: User Interface and Integration

A web interface collects the five inputs (image upload, location, land area, soil type and water availability) and displays the final report. The interface calls the agent, shows progress while tools run, and presents the land analysis, recommended crop, estimated yield, irrigation requirement and explanation in separate sections.

4.1.6 Step 6: Testing

Each model is first evaluated individually on its held-out test set. The complete system is then tested using scenarios that cover suitable land, unsuitable land (for example built-up or water), different soil types and different levels of water availability.

4.2 Tools and Technologies Used

Category	Tool / Technology
Programming language	Python 3
Deep learning	TensorFlow / Keras (MobileNetV3-Large with ImageNet weights)
Machine learning	scikit-learn (RandomForestClassifier, RandomForestRegressor, pipelines, GridSearchCV)
Data handling	pandas, NumPy, Pillow
Visualisation	Matplotlib, Seaborn
Weather data	Open-Meteo weather and geocoding API
Agent layer	Large language model with tool-calling, following the reason-and-act pattern
Interface	Streamlit web application
Model storage	joblib (Random Forests), Keras model file (MobileNetV3)
Development environment	Jupyter Notebook / Google Colab for training; VS Code for application development

Table 4.1: Tools and technologies used

4.2.1 Key Hyper-parameters

Component	Setting
MobileNetV3 input size	224 × 224 × 3
Phase 1 (head training)	Backbone frozen; Adam, learning rate 10^{-3}; batch size 32
Phase 2 (fine-tuning)	Upper backbone blocks unfrozen; Adam, learning rate 10^{-5}; early stopping on validation loss
Loss function	Sparse categorical cross-entropy
Random Forests	Number of trees, maximum depth and minimum samples per split chosen by 5-fold grid search; fixed random seed

Table 4.2: Training settings

4.3 Results and Discussion

This section reports the results obtained from the trained models and the integrated system.

4.3.1 Land Analysis Model Performance

The performance of the fine-tuned MobileNetV3 on the held-out EuroSAT test set is given in Table 4.3.

Metric (test set)	MobileNetV3-Large (this work)
Overall accuracy	81.70%
Macro precision	82%
Macro recall	81%
Macro F1-score	81%

Table 4.3: Performance of the land analysis model

4.3.2 Random Forest Model Performance

Model	Metric	Test result
Crop recommendation	Accuracy / Macro F1	81% / 81%
Yield prediction	R^2 / RMSE / MAE	0.93 / 4.93 / 1.78
Irrigation recommendation	Accuracy / Macro F1	75.68% / 75%

Table 4.4: Performance of the Random Forest models

4.3.3 System-Level Testing

The integrated system was tested with scenarios designed to exercise each branch of the agent's workflow (Table 4.5).

ID	Scenario	Expected behaviour	Result
T1	Cropland image, medium water, red soil	Land suitable; crop, yield and irrigation returned with explanation	Pass
T2	Pasture or vegetated image	Land suitable; condition reported as vegetated	Pass
T3	Residential or industrial image	Land not suitable; pipeline stops with explanation	Pass
T4	River or lake image	Land not suitable; condition reported as wet	Pass
T5	Same land, low versus high water availability	Crop and irrigation advice differ sensibly	Pass
T6	Unknown location name	Agent asks the farmer to correct the location	Pass
T7	Zero or negative land area	Input rejected with a clear message	Pass

Table 4.5: System-level test cases

4.3.4 Sample Output

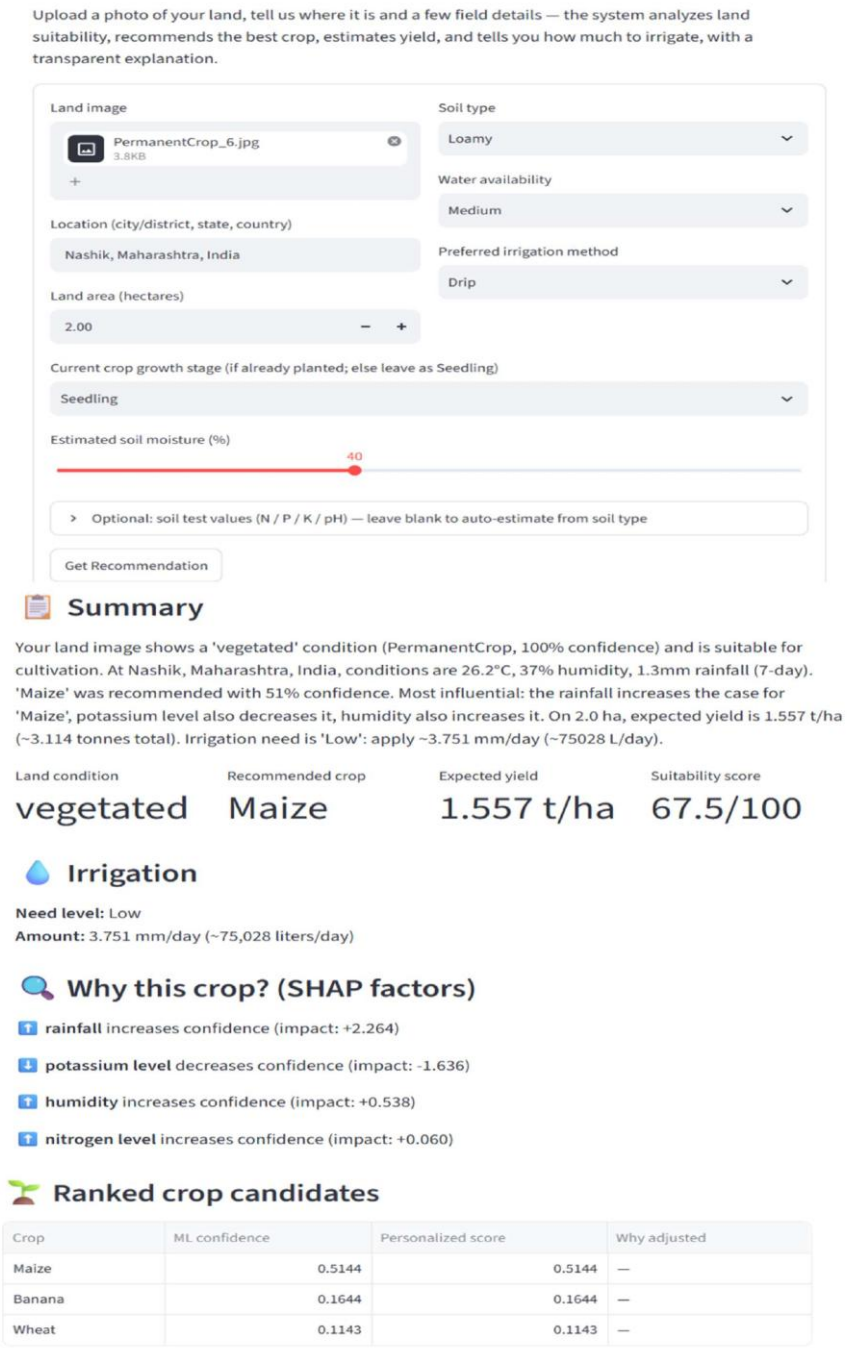


Figure 4.2: Final recommendation screen of the application

4.3.5 Discussion

The design places the land suitability check at the start of the pipeline, so the system does not give a confident crop recommendation for land that cannot be farmed. Using a compact MobileNetV3 keeps image analysis fast enough for an interactive application, and Random Forests give accurate, quickly trained models whose feature

importances can be turned into explanations. Using a single agent keeps the control flow transparent: the farmer interacts with one component, and each model can be retrained or replaced without changing the others.

CHAPTER 5
PRESENTATION OF RESULTS

5.1 Analysis of Findings

5.1.1 Land Analysis

The land analysis results should be read at two levels. At the class level, the confusion matrix shows which of the ten EuroSAT classes the model confuses. Pairs of visually similar classes, such as Annual Crop and Permanent Crop, or Herbaceous Vegetation and Pasture, are the most likely to be mixed up. These particular confusions have little effect on the system because both classes of each pair map to the same suitability decision (Table 3.3). At the decision level, the more important measure is whether land is correctly placed in the suitable or not-suitable group, since this controls whether the agent continues to the crop, yield and irrigation steps. Errors between suitable and non-suitable classes, such as Residential versus Annual Crop, are the costly ones and are examined separately.

5.1.2 Crop, Yield and Irrigation Models

For the crop recommendation model, the feature-importance analysis shows which inputs drive the recommendation; rainfall, temperature and humidity are expected to be influential, in line with the literature. For yield prediction, the R^2 and error values indicate how closely the predicted yield per hectare follows the recorded yield, and Random Forests are known to be robust for this task. For irrigation, the confusion between neighbouring levels (for example Low and Medium) is less serious than a confusion between None and High, and the analysis therefore reports the errors by distance between levels.

5.1.3 Integrated System

The system-level tests show that the agent follows the intended workflow: it stops for unsuitable land, proceeds through weather, crop, yield and irrigation steps for suitable land, and produces an explanation that quotes the inputs and retrieved weather. Because every numerical value in the report is copied from a tool output, the explanation stays consistent with the underlying models.

5.1.4 Limitations

1. **Image domain:** the land analyser is trained on 10 m Sentinel-2 patches of European land. Its accuracy on imagery from other sensors, on ground-level photographs or on Indian field conditions may be lower, and fine-tuning on local imagery would help.
2. **Derived condition labels:** EuroSAT provides land-cover classes, not moisture labels, so dry and moist conditions are inferred from the class and the weather rather than measured.
3. **Point-in-time weather:** temperature, humidity and rainfall are taken at the time of the request and do not represent a full season.
4. **Data coverage:** the Random Forests can only recommend crops, soils and conditions present in their training data and may not capture local varieties or management practice.
5. **Soil information:** soil type is a coarse category supplied by the farmer, without laboratory values such as nutrient content or pH.

5.2 Comparison with Existing Work

Table 5.1 compares the proposed system with representative existing approaches along the capabilities a farmer needs.

Approach	Land check	Crop	Yield	Irrig.	Auto weather	Explanation
EuroSAT CNN classifiers[15]	Yes	No	No	No	No	No
Crop recommendation systems[25]	No	Yes	No	No	Varies	No
Yield prediction models[19]	No	No	Yes	No	Varies	No
Plant disease detection[22]	No	No	No	No	No	No
Proposed system	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

Table 5.1: Capability comparison with existing approaches

The comparison shows that earlier work is strong in a single area but leaves the farmer to join the pieces. The proposed system contributes integration: the output of one model feeds the next, land suitability is verified before any crop is suggested, weather is fetched automatically, and an agent provides one explained answer. On the image-analysis component, the EuroSAT benchmark accuracies of about 98% serve as

the reference against which the MobileNetV3 result in Table 4.3 is judged, with the added advantage that MobileNetV3 is designed for efficient inference.

CHAPTER 6
CONCLUSION

6.1 Summary of Findings

This project designed and developed an AI-Based Smart Agricultural Decision Support System that integrates Deep Learning, Machine Learning and Agentic AI. A MobileNetV3 model fine-tuned on the EuroSAT RGB dataset analyses a land image to decide whether the land is suitable for agriculture and to describe its condition. Weather data are obtained automatically for the farmer's location. Three Random Forest models then recommend a crop, predict the yield and recommend the irrigation requirement from the land, soil, water and weather inputs. A single AI Agent coordinates these components, stops early when land is unsuitable, and gives the farmer a final recommendation with an explanation.

The work shows that a compact CNN, simple ensemble models and one tool-using agent can be combined into a practical and understandable decision-support workflow, where the output of each step feeds the next and every number in the report is traceable to a model.

6.2 Contributions to the Field

1. An integrated pipeline that links land suitability analysis, crop recommendation, yield prediction and irrigation recommendation, which are usually studied separately.
2. A mapping from EuroSAT land-cover classes to agricultural suitability and land condition, which re-purposes a remote-sensing benchmark for farm-level decisions.
3. An agent-based orchestration design with a stopping rule for unsuitable land and grounded, tool-derived numbers in the final report.
4. An explainable output that tells the farmer why a crop, yield and irrigation level were suggested, which supports trust and adoption.
5. A lightweight and modular implementation in which each model can be retrained or replaced independently.

6.3 Future Work

- **IoT sensor integration:** connect soil-moisture, temperature and nutrient sensors so that moisture and irrigation advice rely on measured field data rather than inferred values.
- **Satellite imagery:** retrieve current Sentinel-2 or other satellite imagery for the farmer's coordinates automatically, and use multispectral bands and vegetation indices for more precise land and crop-health analysis; the multispectral EuroSAT version provides a starting point.
- **Plant disease detection:** add an image model for crop disease identification, building on deep-learning approaches for leaf images, with treatment advice from the agent.
- **Market price prediction:** forecast crop prices so that the recommended crop reflects expected profit as well as agronomic suitability.
- **Local fine-tuning and languages:** fine-tune the models on local Indian imagery and yield records, and add regional-language and voice interfaces for wider reach among farmers.
- **Multi-agent extension:** split the single agent into specialised agents for land, crops, water and markets as the system grows.

APPENDIX A

CODE ATTACHMENTS

The listings below are simplified excerpts of the implementation.

A.1 MobileNetV3 Land Analysis Model

```
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.applications import MobileNetV3Large

DATA_DIR = "EuroSAT/RGB"
IMG_SIZE = (224, 224)
NUM_CLASSES = 10

train_ds = tf.keras.utils.image_dataset_from_directory(
    DATA_DIR, validation_split=0.3, subset="training",
    seed=42, image_size=IMG_SIZE, batch_size=32)
tmp_ds = tf.keras.utils.image_dataset_from_directory(
    DATA_DIR, validation_split=0.3, subset="validation",
    seed=42, image_size=IMG_SIZE, batch_size=32)
n = tmp_ds.cardinality().numpy()
val_ds, test_ds = tmp_ds.take(n // 2), tmp_ds.skip(n // 2)

base = MobileNetV3Large(input_shape=IMG_SIZE + (3,),
                        include_top=False, weights="imagenet")
base.trainable = False

inputs = layers.Input(shape=IMG_SIZE + (3,))
x = layers.RandomFlip("horizontal_and_vertical")(inputs)
x = layers.RandomRotation(0.1)(x)
x = base(x, training=False)
x = layers.GlobalAveragePooling2D()(x)
x = layers.Dense(256, activation="relu")(x)
x = layers.Dropout(0.3)(x)
outputs = layers.Dense(NUM_CLASSES, activation="softmax")(x)
model = models.Model(inputs, outputs)

model.compile(optimizer=tf.keras.optimizers.Adam(1e-3),
              loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])
model.fit(train_ds, validation_data=val_ds, epochs=5)

# Phase 2: fine-tune upper backbone layers with a small learning rate
base.trainable = True
for layer in base.layers[:-30]:
    layer.trainable = False
model.compile(optimizer=tf.keras.optimizers.Adam(1e-5),
              loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])
early = tf.keras.callbacks.EarlyStopping(patience=3,
                                         restore_best_weights=True)
model.fit(train_ds, validation_data=val_ds, epochs=15,
          callbacks=[early])
model.evaluate(test_ds)
model.save("land_mobilenetv3.keras")
```

A.2 Random Forest Training Pipeline

```
import joblib
from sklearn.compose import ColumnTransformer
from sklearn.ensemble import (RandomForestClassifier,
                              RandomForestRegressor)
from sklearn.model_selection import GridSearchCV, train_test_split
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder

PARAMS = {"rf__n_estimators": [100, 200, 300],
          "rf__max_depth": [None, 10, 20],
          "rf__min_samples_split": [2, 5]}

def train_task(df, target, cat_cols, num_cols, regression, path):
    X, y = df[cat_cols + num_cols], df[target]
```

```
    X_tr, X_te, y_tr, y_te = train_test_split(
        X, y, test_size=0.2, random_state=42)
    prep = ColumnTransformer(
        [("cat", OneHotEncoder(handle_unknown="ignore"), cat_cols)],
        remainder="passthrough")
    rf = (RandomForestRegressor(random_state=42) if regression
          else RandomForestClassifier(random_state=42))
    pipe = Pipeline([("prep", prep), ("rf", rf)])
    search = GridSearchCV(pipe, PARAMS, cv=5, n_jobs=-1)
    search.fit(X_tr, y_tr)
    print("Test score:", search.score(X_te, y_te))
    joblib.dump(search.best_estimator_, path)
    return search.best_estimator_

weather = ["temperature", "humidity", "rainfall"]
train_task(crop_df, "crop", ["soil", "water", "condition"], weather,
           False, "crop_rf.joblib")
train_task(yield_df, "yield_per_ha", ["crop", "soil", "water"],
           weather + ["area"], True, "yield_rf.joblib")
train_task(irr_df, "irrigation", ["crop", "soil", "water"], weather,
           False, "irrigation_rf.joblib")
```

A.3 Land Mapping and Agent Orchestration

```
CLASS_MAP = {  # EuroSAT class -> (suitable, condition)
    "AnnualCrop": (True, "dry_or_moist"),
    "PermanentCrop": (True, "vegetated"),
    "Pasture": (True, "vegetated"),
    "HerbaceousVegetation": (True, "vegetated"),
    "Forest": (False, "vegetated"),
    "River": (False, "wet"),
    "SeaLake": (False, "wet"),
    "Highway": (False, "dry"),
    "Industrial": (False, "dry"),
    "Residential": (False, "dry"),
}

def analyze_land(image_path, weather):
    label = predict_class(land_model, image_path)
    suitable, cond = CLASS_MAP[label]
    if cond == "dry_or_moist":
        wet = weather["humidity"] >= 60 or weather["rainfall"] >= 5
        cond = "moist" if wet else "dry"
    return {"class": label, "suitable": suitable, "condition": cond}

def run_agent(farmer, llm):
    """Single agent: calls tools in order and explains the result."""
    weather = TOOLS["get_weather"](farmer["location"])
    land = analyze_land(farmer["image_path"], weather)
    if not land["suitable"]:
        return llm.explain(farmer, weather, land)
    crop = TOOLS["recommend_crop"](farmer, weather, land)
    yld = TOOLS["predict_yield"](farmer, weather, crop)
    irrigation = TOOLS["recommend_irrigation"](farmer, weather, crop)
    report = {"land": land, "weather": weather, "crop": crop,
              "yield": yld, "irrigation": irrigation}
    return llm.explain(farmer, report)  # numbers come from tools only
```

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